Zurich ServiceNow AI Platform Capabilities

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CMDB 360/Multisource CMDB

CMDB 360 retains complete history about discovery sources and proposed values, involved in updates of CI attributes. Use CMDB 360 data to track how the CMDB is populated by various discovery sources at the CI attribute level. Also, to revert CI updates from a specific discovery source, or to recompute attribute values using updated reconciliation rules.

Starting with the Utah release, the Multisource CMDB feature is part of the CMDB 360 feature. CMDB 360 provides all the functionality of Multisource CMDB and additional capabilities such as an analytics dashboard, and new query functionality. You can access all of the CMDB 360 capabilities in the CMDB 360 view in CMDB Workspace.

How CMDB 360 works

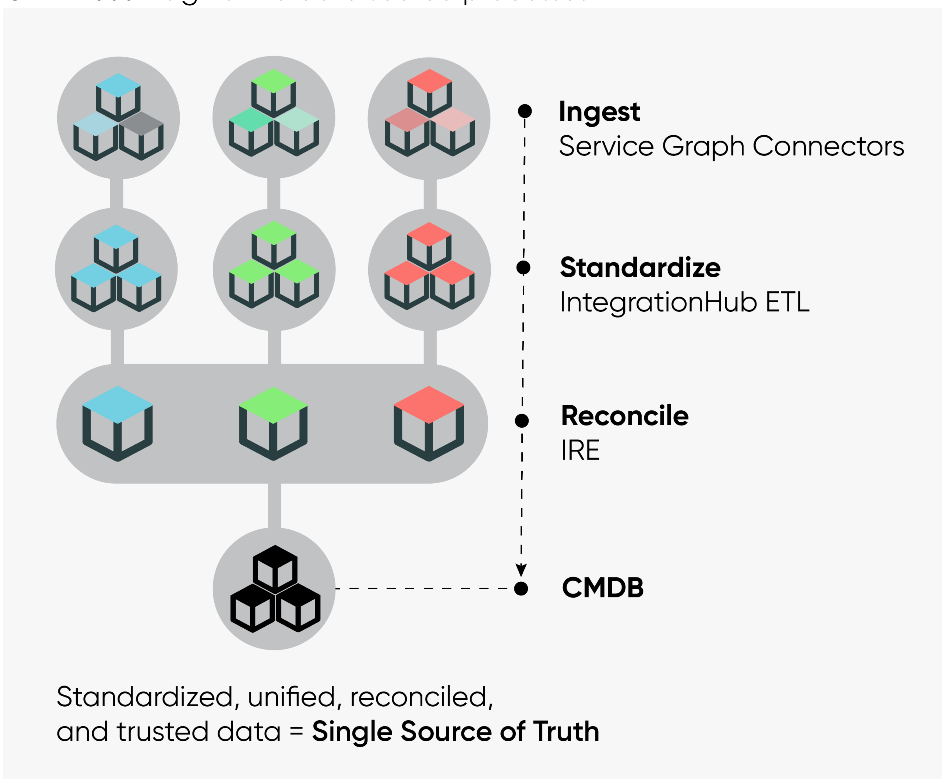
When multiple discovery sources attempt to update the same CI attribute, the Identification and Reconciliation Engine (IRE) uses reconciliation rules to select a single discovery source for the update. Without CMDB 360, details about the lower-priority discovery sources whose values were rejected, are discarded. Also, it is difficult to identify the source of an attribute value without CMDB 360.

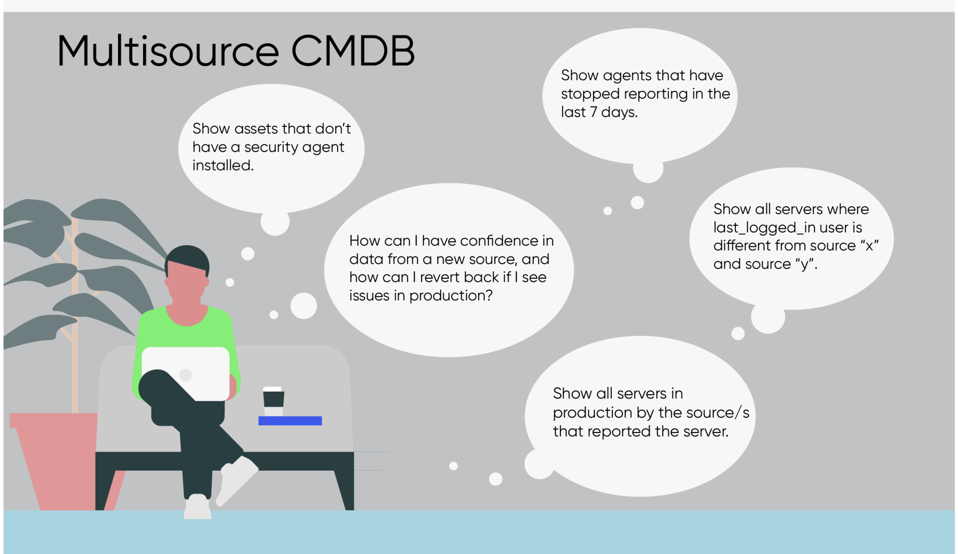
With CMDB 360, the raw details for every discovery source and CI combination are retained for both, discovery sources that were selected for an update and all others that were not. CMDB 360 data, consisting of records for each discovery source and CI combination, is stored in the CMDB MultiSource Data [cmdb_multisource_data] table. You can examine, query, and report on the CMDB 360 data store.

You can optionally exclude classes and their descendents from collecting and processing CMDB 360 data. The CMDB MultiSource Data [cmdb_multisource_data] table doesn't contain data for those excluded classes. For more information, see Exclude classes from CMDB 360.

> **Note:** CMDB 360 supports non-CMDB tables. The widely used term Configuration Item (CI), can also refer to a non-CMDB table record. For information about support for non-CMDB tables, see IRE support for non-CMDB tables.

CMDB 360 insights into data source processes





After data is initially ingested from multiple data sources, several processes are applied to standardize and reconcile the data before it is stored in the CMDB. CMDB 360 provides insights that can help you configure some of these processes.

Using CMDB 360

Use CMDB 360 to:

- Create a dynamic reconciliation rule.
- Control CI updates at the discovery source and CI attribute level.
- Visualize discovery sources of attribute values, at the attribute level.
- Modify reconciliation rules and then recompute CMDB data, reflecting the updated reconciliation rules.
- Revert CMDB data integration from a specific discovery source, if, for example, you realize that the discovery source is not reliable. Recompute CI attribute values, while excluding the discovery source that you want to ignore.
- Validate a new discovery source by comparing its data to data from other discovery sources, which are known to be valid.
- Improve data management, data quality, and operational insights, by querying on CMDB 360 data. Use the CMDB 360 query builder in CMDB Workspace to create queries for CMDB 360 records, discovery sources, and CI records.

Enable and configure CMDB 360

- Activate the ITOM Discovery License (com.snc.itom.discovery.license) plugin.
- Navigate to **All > Configuration > CMDB 360 Properties**. Then, in the CMDB 360 Properties pane ensure that the glide.identification_engine.multisource_enabled (Enables CMDB 360) property is set to **true**.
- Optionally, Exclude classes from CMDB 360.

By default, CMDB 360 tracks discovery source information for CIs from CMDB classes and doesn't collect data for non-CMDB tables. You can independently enable or disable tracking data for CMDB and for non-CMDB classes, using these system properties:

- glide.identification_engine.multisource_cmdb_ci_enabled (Enables capturing CMDB 360 data for CIs from CMDB classes)
- glide.identification_engine.multisource_non_cmdb_ci_enabled (Enables capturing CMDB 360 data for CIs from non-CMDB classes

Report on CMDB 360 data

Use the CMDB 360 view in CMDB Workspace to gain insights into the CMDB 360 data store. Build reports that, for example, do the followings:

- Find CIs not reported by any discovery source.
- Find discovery sources populating data in your CMDB.
- Compare attribute values across discovery sources.
- Compare attribute values between CMDB and other discovery source.
- Limit reports for CMDB 360 data, to a specific application service, technical service, or a CMDB group.

See Sample CMDB 360/Multisource CMDB queries for more details.

Visualize CMDB 360 data

CMDB 360 is highly verbose in the user interface:

- On the Reconciliation Rules page in CI Class Manager, click the **Preview Data** tab to see per attribute, discovery sources that are authorized to update that attribute, in precedence order.
- On a CI form, click the **CMDB 360 Data Preview** related link to see per CI attribute, current value in the CMDB and incoming values from other discovery sources.

Logging

Enable logging for CMDB 360 by adding and enabling the system property glide.cmdb.logger.source.cmdb_multisource. In the Log [syslog] table, search for entries in which `source="cmdb_multisource"`.

CMDB 360 experience in CMDB Workspace

The CMDB 360 view provides aggregations and analysis of CMDB 360 data which you can use to track activities and identify Potential issues of discovery sources. You can also create different types of your own queries and associated schedules and reports to explore CMDB data.

Use the CMDB 360 view in CMDB Workspace to access all of the CMDB 360 capabilities. For information about all CMDB 360 dashboard settings, see Configure the CMDB 360 dashboard.

> **Note:** Most cards on the CMDB 360 dashboard support non-CMDB tables in their aggregation, or can be configured to provide support. However, the CIs not reported by discovery sources card, for example, doesn't apply to non-CMDB tables. Creating queries for non-CMDB tables is also supported. For information about support for non-CMDB tables, see IRE support for non-CMDB tables.

Access

Requirements:

- Role requirement: sn_cmdb_user (CMDB user) or any role containing sn_cmdb_user
- Additional requirement: Enable and configure CMDB 360

To access the CMDB 360 view in the CMDB Workspace, navigate to **Workspaces > CMDB Workspace**. In the CMDB Workspace menu bar, select **CMDB 360**.

Potential issues

Cards on the Potential issues tile show details about CIs with discovery sources that are incorrectly reporting on the CIs.

CIs not reported by discovery sources

Lists CIs that are discovered by multiple discovery sources, but one or more of the discovery sources has stopped reporting within a specified number of days. The Number of days since CIs were last discovered by a discovery source dashboard setting is used in the card's aggregation.

Drill down on this card to see a list view of the CIs for the card and the specific discovery source per CI that is no longer reporting.

Example: You configure 7 days for the setting. A Linux server CI named backup-linux.sea.com is reported by these discovery sources within the specified number of days:

- ServiceNow - today.
- ServiceWatch - 4 days ago.
- AgentClientCollector - 8 days ago.

In this scenario, the CI shows on the CIs not reported by discovery sources card, since AgentClientCollector reported over seven days ago.

Data mismatch

Lists CIs for which different discovery sources are reporting different values. Attributes are considered mismatched when different discovery sources report different values for the attribute. CIs that appear when you drill down on this card can reveal issues with the individual CI, or your reconciliation rules.

The specific records that appear in the drilled-down list view, depend on the Data mismatch dashboard settings.

In the Data mismatch records list view, select an item in a Source column to access the Multisource Data Preview page with details about all discovery sources with values for the attribute. On the Multisource Data Preview page you can use the Search Attributes box to search for specific attributes and also choose one of the following options:

- **All attributes**: Shows all attributes of the selected class, regardless of whether they have a value in the CMDB.
- **Attributes with CMDB values**: Shows only attributes of the selected class, with a value in the CMDB.
- **Attributes with multisource data**: Shows all attributes of the selected class, and for each attribute shows current CMDB value and any other discovery sources with a value for the attribute.

Regardless of the option that you choose on the Multisource Data Preview page, a discovery source value that was used for the current

CMDB value – is highlighted in green. These values are identical to the CMDB value of the attribute.

Saved queries

The Saved queries card shows up to 20 of your CMDB 360 queries. You can use the card to edit and run those queries, or create new queries. Saved queries are sorted on both the card and list view by the most recently created or updated queries.

If your instance has been upgraded from an instance that contained Multisource Report Builder queries, then those queries appear on the Saved queries card.

- Click a saved query to modify or view it before running.
- Click a query's action icon and then select **Run**.
- Click **View All Queries** to see all the Saved queries where you can examine or run a query.
- Click **Create Query** to create a query of one of the following types:
  - Get Records: Creates a query that you can use to explore your CMDB 360 data. It queries your CMDB 360 for CIs matching your criteria that are reported by specified discovery sources.
  - Find Gap: Creates a query that you can use to analyze gaps in discovery sources reporting your CMDB 360 data. It queries discovery sources that report CIs against discovery sources that don't report those same CIs.
  - Compare Attribute Values: Creates a query that you can use to identify CIs with attribute values that differ across multiple discovery sources or against the CMDB. It queries at least two discovery sources and/or the CMDB for CIs that match your criteria.
- Schedule a CMDB 360 query for a report to run that query on a regular basis. Scheduling a query enables you to create a CMDB 360 report that integrates the query results with the platform Reporting feature.

Discovery sources

Cards in the Discovery sources tile show aggregated counts for your discovery sources. This tile also includes coverage cards showing a breakdown of CIs per the number of discovery sources reporting those CIs.

Number of discovery sources

The total number of discovery sources that report CMDB 360 data.

Total CMDB 360 records

The total number of raw CMDB 360 records in the CMDB 360 data store that contains records for each discovery source report, per each CI attribute.

Reconciled CIs

The total number of unique CIs created in the CMDB after processing incoming data from all discovery sources, including after reconciling data from multiple discovery sources for the same CIs. For more information, see CMDB Identification and Reconciliation.

Discovery source overviewt

The distribution of CMDB 360 CIs across all reporting discovery sources.

Coverage: CIs with a single discovery source

Shows CIs that are only reported by a single discovery source, with a breakdown by the discovery source reporting those CIs. The number on the chart is the actual total number of CIs with a single discovery source, regardless of any dashboard settings. The number of records in list views that appear when you drill down the chart, depends on dashboard settings and can be only a subset of that total.

Coverage: CIs by number of discovery sources

Shows CIs reported by multiple discovery sources, grouped by the number of discovery sources that are reporting those CIs.

You can use the following options to filter the calculated records in the CMDB 360 data store that apply to all cards in the Discovery sources tile:

- All (default): Includes records from both CMDB and non-CMDB tables.
- CMDB CI: Includes records only from CMDB classes.

- Principal Class: Includes records only from principal classes. For information about defining a class as principal and therefore including that class in the Principal Class filter, see Create a CI class.

Various settings apply only to the coverage charts. They determine the scope and type of data that is included in the coverage charts and can greatly affect the performance of the associated calculation job:

- The dashboard Coverage settings determine the mixture of classes for the records in drill-down list views. For information about these settings and their effect, see Coverage cards.
- The system properties sn_cmdb_ws.ms.calculate_cmdb_only and glide.identification_engine.multisource_non_cmdb_ci_enabled determine if both CMDB and non-CMDB or CMDB only data, is included in the coverage charts. For more information about these system properties, see Components related to CMDB 360.

  Note: If the sn_cmdb_ws.ms.calculate_cmdb_only is true or glide.identification_engine.multisource_non_cmdb_ci_enabled is false, then both 'All' and 'CMDB CI' filters show CMDB data only (non-CMDB data is ignored).

- The sn_cmdb_ws.ms.report_class_ci_count_max_threshold system property and the CMDB 360 Analytics Skipped Class [sn_cmdb_ws_ms_skip_class] table, together, determine class inclusion in the coverage charts.

  If the record count for a class (CMDB or non-CMDB) in the CMDB 360 Data [cmdb_multisource_data] table exceeds the value set by the sn_cmdb_ws.ms.report_class_ci_count_max_threshold system property, then a record is created in the CMDB 360 Analytics Skipped Class [sn_cmdb_ws_ms_skip_class] table and by default, the class is skipped in future calculations for the coverage charts. You can include a skipped class in the coverage charts by setting the Override class CI records population column of a skipped class in the sn_cmdb_ws_ms_skip_class table, to 'true'.

Note: Some classes, when included or excluded from the coverage charts, can greatly impact the overall performance of the chart calculations. For more information, see Components related to CMDB 360.

Configure the CMDB 360 dashboard

Configure settings on the CMDB 360 of the CMDB Workspace to determine how your CMDB 360 data is analyzed and aggregated. These settings affect the data that appears on the cards and the records shown when you drill down on those cards on your CMDB 360 dashboard.

About this task

For detailed information about CMDB 360-related components, such as system properties, scheduled jobs, and tables, that are installed with CMDB Workspace, see Components installed with CMDB Workspace.

Before you begin

Role required: cmdb_ms_admin

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace menu bar, select **CMDB 360**.
3. Select **Settings**.
4. Configure Global settings.
   The **Maximum number of records in a list view** setting determines the maximum number of records that can show when you drill down on the cards in the CMDB 360 dashboard (drill-down set). If the total number of returned records is greater than the specified setting value, the dashboard trims the list view output according to this setting and the settings of individual cards.
   You can use this setting to limit the number of records that CMDB 360 must process. The setting applies to these cards:
   - CIs not reported by discovery sources

- Data mismatch
- Coverage: CIs with a single discovery source
- Coverage: CIs by number of discovery sources

The CMDB 360 dashboard defaults this value to 100,000.

5. Configure Potential issues settings.
These settings affect the calculations for cards on the CMDB 360 view/Potential Issues tile and the list of CIs that appear when you drill down on those cards.

    a. Configure CIs not reported by discovery sources.
    The **Number of days since CIs were last discovered by a discovery source** setting determines the number of days used in the calculation of the CIs not reported by discovery sources card. The card shows CIs that are discoverable by multiple sources, but at least one discovery source hasn't reported on that CI in the specified number of days.

    b. Configure Data mismatch.

    These settings determine the classes of CIs included in the calculation and in the drill-down list views of the Data mismatch card. These settings determine which classes to use for the card and the relative weight (%) of each of those classes in the card calculations.

    Using these settings, add the classes for which you are interested in seeing CMDB 360 mismatch data. Regardless of the classes that you add in the Data mismatch settings, the 'Maximum number of records in a list view' setting, is always in effect when you drill down the card (100,000 by default).

Setting	Description
Automatic data weights	Evenly distributes weights among any added classes. The global setting of 'Maximum number of records in a list view' is evenly divided

Setting	Description
	between all added classes so that each class can contribute an equal number of records to the drill-down set. For example, if you added four classes, the weight of each class = 25. Therefore, the system adds to the set of drill-down records, up to 25,000 CMDB 360 records (25% of 100,000) of each added class.
Manual data weights	Custom weight (%) for each added class. The global setting of ‘Maximum number of records in a list view’ is divided between the added classes, according to the specified class weights. For example, if you added two classes, class A weight = 25 and class B weight = 75. The system adds to the set of drill-down records, up to 25,000 CMDB 360 records (25% of 100,000) of class A, and up to 75,000 CMDB 360 records (75% of 100,000) of class B.
Select CI classes you want to include in the calculation	Specify the CI classes that you want to check for attribute mismatches. CI classes you specify also include any child classes. Attributes are considered mismatched when different

Setting	Description
	discovery sources report different values for the attribute.
Show CIs where EVERY attribute doesn't match	Select to include only CIs with a mismatch between discovery sources for every attribute that you specify.
Show CIs where ANY attribute doesn't match	Select to only include CIs with a mismatch between discovery sources for any attribute you specify.
Select an attribute	Specify the attributes that you want to check for mismatches.

6. Configure Coverage settings.

   These settings affect the calculations related to the coverage cards ('Coverage: CIs with a single discovery source' and 'Coverage: CIs by number of discovery sources') in the Discovery Sources tile on the CMDB 360 view. These settings determine the mixture of classes in the drill-down set of records.

   Using these settings, you can give priority to classes for which you are most interested in seeing CMDB 360 data in the Coverage charts. The number of records that you can drill down to in the Coverage charts, is limited by the global setting 'Maximum number of records in a list view' (100,000 by default). Regardless of your settings and class priorities, the 'Maximum number of records in a list view' setting, is always in effect.

   By default, the classes of the records in the drill-down set are random. You can add classes in this Coverage settings to ensure that your preferred classes have priority in being included in the drill-down set of records. You can prioritize any number of classes and assign a weight (%) for each of those classes within the total number of drill-down records.

Setting	Description
Automatic data weights	Evenly distributes weights among any added classes. The global setting of 'Maximum number of records in a list view' is evenly divided between all added classes so that each class can contribute an equal number of records to the drill-down set. For example, if you added four classes, the weight of each class = 25. Therefore, the system adds to the set of drill-down records, up to 25,000 CMDB 360 records (25% of 100,000) of each added class.
Manual data weights	Custom weight (%) for each added class. The global setting of 'Maximum number of records in a list view' is divided between the added classes, according to the specified class weights. For example, if you added two classes, class A weight = 25 and class B weight = 75. The system adds to the set of drill-down records, up to 25,000 CMDB 360 records (25% of 100,000) of class A, and up to 75,000 CMDB 360 records (75% of 100,000) of class B.
Select CI classes you want to include in the calculation	Prioritized classes for which you are most interested in

Setting	Description
	seeing CMDB 360 data in the Coverage charts (‘CIs with a single discovery source’ and ‘CIs by number of discovery sources’). CI classes that you specify also include any child classes.

Using the Coverage settings, the system calculates an allowance of records per class, in the drill-down set of records. If the number of actual records per added class, is less than the computed allowance for that class, then the system adds records of random classes, up to the computed allowance for the class.

7. Select **Save**.

Create a Get Records query

Create a Get Records query from the CMDB 360 dashboard of your CMDB Workspace to help you explore your existing CMDB 360 data.

Before you begin

Role required: sn_cmdb_user and either cmdb_ms_admin or cmdb_ms_editor

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace menu bar, select **CMDB 360**.
3. On the Saved Queries tile, select **Create Query.**
4. Select **I want to get CMDB 360 data**.
5. Select the CI classes to include in the query.

You can select a class to open the condition builder. Use the condition builder to specify conditions that must be met for each class. Use **And** or **Or** to specify multiple conditions.

Select **All Classes** if you want to include all CI classes without conditions.

6. Select **Continue**.
7. Select discovery sources to query on.
   The query retrieves CMDB 360 data that originates from the discovery sources you specify.

   You can leave the Select discovery sources prompt empty to retrieve data for all discovery sources.
8. Select **Continue**.
9. On the form, select the options:

Results Layout form

Field	Description
Show unique CMDB 360 records	Select if you want to see only unique CMDB 360 records. Records for the same CIs from different discovery sources are consolidated.
Show CI records by discovery source	Select if you want to see records for each CI and discovery source pair.
Limit results to	Limits the query results to CIs that belong to a service or CMDB group. When you select **Application Services**, **Technical Services**, or **CMDB Groups**, a prompt appears. You can use the prompt to specify the service or group that you want the query to filter for.

10. Select **Continue**.
11. Enter a name and description for your query.
12. Select **Save** .

What to do next

Run the query at least once if you want to create a schedule or report.

On the CMDB 360 Query Results page:

- If the number of results exceeds the number of results appearing on the page:
  - Select **Load More Results**: To show the next page of results. The number of results that appear on each result page is specified by the glide.identification_engine.multisource.query.batch.limit system property (100 items by default).
  - Select **Load All Results**: To show all results, up to the limit specified by the glide.identification_engine.multisource.query.max.limit system property (10000 by default).
  - Select a CMDB 360 Source link to easily access preview data of a source and see more details.
- You can select **Create Schedule** to set up a schedule that runs your query on a regular basis. Scheduling your query enables you to use the query results in reports you create.
- After creating a schedule, you can select **Create Report** to configure a report that you can manage using Reporting capabilities.
- On the Query Results page, access a record to view further details. For example, select a link in the Primary Record column, and then in the CI Details page, select **View CMDB 360 Data**.

Create a Find Gap query

Create a Find Gap query from the CMDB 360 dashboard of your CMDB Workspace to help you find CIs that are not being reported by a discovery source.

Before you begin

Role required: sn_cmdb_user and either cmdb_ms_admin or cmdb_ms_editor

About this task

Gaps in discovery source reporting occur when at least one discovery source reports on the CI and another doesn't, enabling you to identify discovery sources that might have an issue.

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace menu bar, select **CMDB 360**.
3. Select **Create Query.**
4. Select **I want to find gaps in data between discovery sources**.
5. Select the CI classes that you want to check for gaps.
6. Select **Continue**.
7. Select non-reporting discovery sources.
   A non-reporting discovery source is a discovery source that doesn't report CIs as expected. Multiple non-reporting discovery sources have an OR condition with each other.
8. Select one or two reporting discovery sources.
   These discovery sources report the CI as expected and act as a baseline for identifying the gap in reporting.

   You can leave the **Select reporting discovery sources for gap tracking** prompt empty to include all discovery sources in the query.
9. Select **Continue**.
10. On the form, select the options:

Results Layout form

Field	Description
Show unique CMDB 360 records	Select if you want to see only unique CMDB 360 records. Records for the same CIs from different discovery sources are consolidated.
Show CI records by discovery source	Select if you want to see records for each CI and discovery source pair.
Limit results to	Limits the query results to CIs that belong to a service or CMDB group. When you select **Application Services**, **Technical Services**, or **CMDB Groups**, a prompt appears where you can specify the service or CMDB group that you want the query to filter for.

11. Select **Continue**.
12. Enter a name and description for your query.
13. Select **Save** .

What to do next

Run the query at least once if you want to create a schedule.

On the CMDB 360 Query Results page:

- If the number of results exceeds the number of results appearing on the page:
  - Select **Load More Results**: To show the next page of results. The number of results that appear on each result page is specified by the glide.identification_engine.multisource.query.batch.limit system property (100 items by default).

  - Select **Load All Results**: To show all results, up to the limit specified by the glide.identification_engine.multisource.query.max.limit system property (10000 by default).
  - Select a CMDB 360 Source link to easily access preview data of a source and see more details.
- You can select **Create Schedule** to set up a schedule that runs your query on a regular basis. Scheduling your query enables you to use the query results in reports you create.
- After creating a schedule, you can select **Create Report** to configure a report that you can manage using Reporting capabilities.
- On the Query Results page, access a record to view further details.

Create a Compare Attribute Values query

Create a Compare Attribute Values query from the CMDB 360 dashboard of your CMDB Workspace to help you find CIs with mismatched attribute values between discovery sources.

Before you begin

Role required: sn_cmdb_user and either cmdb_ms_admin or cmdb_ms_editor.

About this task

The query enables you to determine if there's an issue with how a discovery source reports a CI.

The Compare Attribute Values query compares CIs from different discovery sources for mismatched values. Mismatches occur when a discovery source reports attribute values for a CI that are different from values reported by other discovery sources or the CMDB. You can use the query to identify these CIs and reconcile the attribute values, or fix any issues with the discovery sources.

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.

2. In the CMDB Workspace menu bar, select **CMDB 360**.
3. Select **Create Query**.
4. Select **I want to compare attribute values between discovery sources or against the CMDB**.
5. Select a CI class that you want to compare attribute values for.
   You can select a selected class to open the condition builder. Use the condition builder to specify conditions that must be met for each class. Use **And** or **Or** to specify multiple conditions.
6. Select **Continue**.
7. On the Attributes to compare form, select options:

Option	Description
Any attribute doesn't match	Select if you want to retrieve CIs where there's a mismatch with any specified attribute values between the discovery sources.
Every attribute doesn't match	Select if you want to retrieve CIs where there's a mismatch with every specified attribute value between the discovery sources.
Select attributes to compare	Specify the attributes that you want to compare for mismatched values.

8. Select **Continue**.
9. Select the discovery sources that you want to compare attribute values for.

Discovery sources form

Field	Description
Compare to CMDB	Select if you want to compare the specified attribute values against your CI attributes

Field	Description
	recorded in the CMDB. When you compare against the CMDB, you only need one discovery source.
Select discovery sources	The discovery sources that you want to compare. Select at least two.
Limit results to	Limits the query results to CIs that belong to a service or CMDB group. When you select **Application Services**, **Technical Services**, or **CMDB Groups**, a prompt appears. You can use the prompt to specify the service or group that you want the query to filter for.

10. Select **Continue**.
11. Enter a name and description for your query.
12. Select **Save** .

What to do next

Run the query at least once if you want to create a schedule.

On the CMDB 360 Query Results page:

- If the number of results exceeds the number of results appearing on the page:
  - Select **Load More Results**: To show the next page of results. The number of results that appear on each result page is specified by the glide.identification_engine.multisource.query.batch.limit system property (100 items by default).
  - Select **Load All Results**: To show all results, up to the limit specified by the glide.identification_engine.multisource.query.max.limit system property (10000 by default).

  - Select a CMDB 360 Source link to easily access preview data of a source and see more details.
- You can select **Create Schedule** to set up a schedule that runs your query on a regular basis. Scheduling your query enables you to use the query results in reports you create.
- After creating a schedule, you can select **Create Report** to configure a report that you can manage using Reporting capabilities.
- On the Query Results page, access a record to view further details.

Schedule a CMDB 360 query for a report

Set up a schedule to regularly query for CMDB 360 data. Use scheduled queries to provide CMDB 360 data to reports you create, which can provide insight into how discovery sources populate the CMDB and the reliability of those discovery sources.

Before you begin

Ensure that you run the CMDB 360 query at least once.

Role required: sn_cmdb_user and either cmdb_ms_admin or cmdb_ms_editor.

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace menu bar, select **CMDB 360**.
3. On the Saved queries tile, create or access a CMDB 360 query.
   If you created a new query, you must run the query at least once before you can select **Create Schedule** on the query results page.
4. Select **Schedule query** on the Results Layout page of the query.
   To create a schedule for the Compare attributes values query, select **Schedule query** on the Discovery Sources page.
5. Specify a **Run** frequency and time you want to schedule the query to run.
   When you select Weekly or Monthly, you must also select a day of the week or calendar day, respectively.

6. Select **Save**.

What to do next

Create a CMDB 360 report to integrate CMDB 360 query results with platform Reporting capabilities. Each run of the query automatically updates the generated report.

Related tasks

- Create a Get Records query
- Create a Find Gap query
- Create a Compare Attribute Values query

Exclude classes from CMDB 360

Prevent CMDB 360 from collecting, storing, and analyzing data for classes for which it isn't needed. CMDB 360 processes large amounts of data. Therefore, excluding classes can help improve the performance of the Multisource Dashboard Analytics Population scheduled job and of CMDB 360 queries.

Before you begin

Role required: cmdb_ms_admin (automatically included in the CMDB Admin role)

About this task

To exclude classes from CMDB 360, you must directly manage records in the CMDB 360 denied classes [cmdb_multisource_deny_class] table. For each class that you want to exclude, add an active record for that class. Excluding a class from CMDB 360 also automatically excludes any of its descending classes. In the base system, the CMDB 360 denied classes table is populated by default with some classes, such as dscy_net_base and discovery_net_base.

Procedure

1. Select **All** and then, in the Filter navigator, enter `cmdb_multisource_deny_class.list` to open the CMDB 360 denied classes table.
2. Select **New** and then fill out the form.
    - Select the class to exclude (with its descendents) in the field labeled **Deny this class and all extended classes from this class to store CMDB 360 data**.
    - Ensure that **Active** is selected. If **Active** isn't selected, then the specified class isn't excluded from CMDB 360.
    - Select **Submit**.

Result

Existing data for the specified class and its descending classes will be gradually removed from the CMDB 360 Data [cmdb_multisource_data] table using record cleaners, without any impact to the system.

What to do next

You can deselect **Active** or completely delete a record for a class that you want to include in CMDB 360 data collection.

Recompute CI attribute values

Modify reconciliation rules, or exclude a discovery source which is found to be invalid. Then, use the updated reconciliation rules in recomputing CI attribute values, for which those reconciliation rules or discovery source are applicable to.

Before you begin

Enable and configure CMDB 360.

If you want to recompute to apply updated reconciliation rules, then you must first update the reconciliation rules.

Role required: itil_admin or sn_cmdb_admin

About this task

CMDB 360 automatically generates a recompute task for each recompute that you submit. If you submit multiple recomputes, a recompute task is generated for each operation, but only one task runs at any given time. To list all recompute tasks, enter `cmdb_multisource_recomp_task.list` in the left navigation search box.

There is a maximum number of records that can be included in a single recompute operation. This number is specified by the system property glide.identification_engine.multisource.recompute.max.ci.limit (100,000 by default).

Note:

- Recompute skips CIs which are reported by multiple discovery sources, but with different class names.
- Recomputing CI attribute values is not supported with non-CMDB tables.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select a class from the CI Classes hierarchy list.
3. In the left-side pane, expand **Class Info**, and select **Reconciliation Rules**.
4. On the Reconciliation Rules page, click **Recompute**.
5. Select the **Recompute Type**.

    - **Replace values from the specified discovery source with value from the discovery source next in priority, according to reconciliation rules**: Recompute attribute values for the class CIs, applying priorities specified in reconciliation rules and while excluding the specified **Discovery Source**. Records for the discovery source you are excluding, are also removed from the CMDB 360 data store.

This operation applies to data that exists in the CMDB. If reconciliation rules remain in effect for the discovery source that you have excluded, then future data from that discovery source, can populate the CMDB.

- **Apply updated reconciliation rules**: Recompute attribute values for the class CIs, applying the updated reconciliation rules which are now in effect.

6. Select the **Recompute Scope**.
   The scopes grow from one option to the next:
   - **Recompute only CIs of this class**: Basic scope of CIs to recompute.
   - **Recompute CIs of this and derived classes**: Expand the basic scope to include CIs from derived classes.
   - **Recompute CIs of this and derived classes, along with selected related items**: Expand the previous scope to also include CIs from specified related items. Select the related items to include in the recompute.
7. Select the **Delete action** for CIs for which the excluded discovery source, is the only discovery source.
   - **Delete record**: Delete the CI record from CMDB.
   - **Set record attributes to custom value**: Set a specified CI attribute to a custom value to remove the CI from regular operations without deleting the CI. For example, set the Operational status attribute to Retired.
8. Select **Next**.
9. On the Review page, carefully review the counts for the affected CIs to ensure that all record counts are as you expect.
10. Select **Back** to adjust any settings for the recompute.
11. Select **Recompute**.

What to do next

You can do any of the followings:

-

  In the status message that appears for the recompute operation, click the link to see the CMDB 360 Recompute Task for more details. The Recompute Task shows the progress and status of the recompute operation.

  You can abort the recompute by setting **Status** to **Closed Incomplete** and selecting **Update**. You can then set **Status** back to **Work in progress** to resume recompute from where it was aborted.
- Enter `cmdb_multisource_recomp_task.list` in the left navigation search box, to see the status and progress of all recompute tasks.

Multisource Report Builder (legacy)

Improve CMDB data management by querying and reporting on Multisource CMDB data. Use the Multisource Report Builder to gain insights about how discovery sources are populating the CMDB and their reliability. You can then adjust reconciliation rules to improve the quality of CMDB data, if needed.

CMDB 360 in CMDB Workspace

Starting with the Zurich release, the Multisource CMDB feature is part of the CMDB 360 feature which is accessible in the CMDB Workspace. Create, view, modify, schedule, create reports, and run CMDB 360 queries using the CMDB 360 query builder in the CMDB Workspace store app. Use the CMDB 360 query builder to create queries of the following types:

- Get Records: Queries your discovery sources for CIs that match your criteria.
- Find Gap: Queries for gaps in discovery sources reporting your CMDB 360 data. Queries discovery sources that report CIs against discovery sources that don't report those same CIs.
- Compare Attribute Values: Queries for CIs with attribute values that differ across multiple discovery sources or against the CMDB. Queries at

least two discovery sources and/or the CMDB for CIs that match your criteria.

Legacy Multisource Report Builder

Instead of using the CMDB 360 query builder in CMDB Workspace, you can still use the legacy Multisource Report Builder as described in this topic.

After you create a Multisource query in the Multisource Report Builder, you can run the query to see the results. You can then also create a Multisource report that integrates the Multisource query results with the platform Reporting capabilities. High level steps for creating a Multisource report:

1. Create a query, then save and run it.
2. Create a schedule for the query.
3. Create a Multisource report that is based on the Multisource query.

You can create queries that find:

- All the discovery sources populating data in your CMDB.
- CIs not reported by any discovery source.
- All CIs discovered by one discovery source, but not by another discovery source.

You can create other queries that show differences in CI attribute values between multiple data sources, while being compared to the CMDB:

- Show how an attribute value is different between a discovery source and the current CMDB record. For example, find Hardware CIs with different location than what SCCM reports.
- Show how an attribute is different between SourceA, SourceB, and SourceC. For example, show all Computer CIs where RAM is different between SCCM, ServiceWatch, and CMDB.

You can show query results by CI records, Multisource CMDB data records, or discovery sources. You can also limit the report results to CIs within a specific application service, technical service, or a CMDB group.

Create a Multisource query

Query the Multisource CMDB data to gain insights about how discovery sources are populating the CMDB, and then use that query to create a Multisource data report.

Before you begin

Enable and configure CMDB 360.

Role required: cmdb_ms_editor

About this task

The Multisource Report Builder page updates dynamically as you set fields. Therefore, some of the fields that are described in the steps below might not appear.

Procedure

1. Navigate to **All > Configuration > Multisource Report Builder**.
2. On the Multisource Report Builder page, select a query to edit or run, or click **New**.
3. Enter **Name** and **Description** for the query.
4. Select the **Result type** for the query.
    - **CI records**: Results show unique CIs from the Multisource CMDB data store.
    - **Multisource data records**: Results show all entries of CI/discovery source combinations from Multisource CMDB data.
    - **Discovery sources**: Results are grouped by discovery sources that match the query criteria.
5. Select **Only show difference** to show differences in CI attribute values and then select the **Type of difference**.

- **Between CMDB record and discovery source**: Show differences in attribute values between the CMDB data and the specified discovery sources.
- **Between discovery sources**: Show differences in attribute values between specified discovery sources based on Multisource data.

6. Select the **Class** to apply the query to, or select **All Classes** to apply the query to all classes. Use the condition builder to specify conditions that must be met for the class. Use **And** or **Or** to specify multiple conditions.
7. Use the list collector to select one or more **Discovery source** items to query on.
8. Set **Field to compare** to the class attributes for which to show differences.
Use the OR and AND operators to compare based on multiple attributes. The list of attributes is a pre-populated subset of the class attributes, to which you cannot add or remove items.
9. Set **Limit results to** to limit the query results to CIs that belong to a specific application service, technical service, or a CMDB group.
10. Click **Save** and then click **Run**.

What to do next

- On the CMDB Multisource Query Results page:
  - If the number of results exceeds the number of results appearing on the page:
    - Click **Load More Results**: To show the next page of results. The number of results that appear on each result page is specified by the glide.identification_engine.multisource.query.batch.limit system property (100 items by default).
    - Click **Load All Results**: To show all results, up to the limit specified by the glide.identification_engine.multisource.query.max.limit system property (10000 by default).
    - Click a Multisource CMDB value link or a CI value link to access the respective records and see more details.

- In the Configuration Item column, click a CI link to open the CI form. In the Related Links section on the CI form, click the **Multisource Data** tab to show Multsource data, such as discovery sources, related to the CI.

- Create a schedule for the query and ensure that the schedule runs at least once. This step is required for creating a Multisource report.

  Note: Initially, after creating a query, both the **Create Schedule** and the **Create Report** buttons are grayed out. Only after saving the query, you can create and run a schedule for the query, and only then you can create a report that is based on the query.

Schedule a Multisource query

After saving and running a Multisource query, create a schedule for the query to run automatically on a set schedule. Query results are stored in a results table and you can configure email addresses for the results to be sent to or include the results in CMDB dashboards.

Before you begin

The query for the schedule must be already saved and run at least one time.

Role required: cmdb_ms_user

Procedure

1. Navigate to **All > Configuration > Multisource Report Builder**.
2. On the Multisource Report Builder page, select the query that you want to create a schedule for.
3. On the Multisource Report form, click **Create Schedule**.
4. On the Scheduled Email of Multisource Report Builders, click **New**.
5. Fill in the Scheduled Email of Multisource Report Builders form.

Field	Description
Query	Saved query that was created in Multisource Report Builder.
Users	Users to email the query results to.
Groups	User groups to email the query results to.
Run Time	Frequency and time to run the query automatically. When you set **Run** to **On Demand**, the query runs only when you run it manually.
Email addresses	More ad-hoc email addresses to email the query results to.
Subject	Text that will appear as the subject of the email with the query results.
Introductory message	Text that is included in the body of the email with the query results.
Type	Type of the file with the query results, which will be attached to the email.
Zip output	Enables zipping of the results file.
Conditional	Enable a condition for running the query and specify the condition in the **Condition** field. If

Field	Description
	the specified condition isn't met, the query doesn't run.
Condition	Condition that must be met for the query to run (Java script). Appears only if **Conditional** is selected.
Omit if no records	Disable sending an email for a query run that returns no results.

Note: When using update sets to port Multisource schedules from a non-production to a production environment, check the **Users** and **Groups** settings in the schedule. Any user or group that doesn't exist in the production environment, and which needs to receive the query results, must be re-added in either of the following ways:

- Manually created in the production environment. In this case, you must also remove the invalid user or group in the production environment (ported from non-production environment) from any schedules and add the new user or group instead.
- Explicitly ported from the non-production to production environment.

6. Click **Submit**.

What to do next

- If **Run** is set to **On Demand** in a schedule, or if you need to run a query randomly even if it has a recurring schedule, you can manually run that query as follows:

  1. Navigate to **All > Configuration > Multisource Report Schedules**.
  2. In the Scheduled Email of Multisource Report Builders list view, select the query that you want to run.

3. On the Scheduled Email of Multisource Report Builder form, click **Execute Now**.

- Create a Multisource report for the query, that integrates the Multisource query results with the platform Reporting feature.

Create a report based on a Multisource (CMDB 360) query

After creating, saving, running, and scheduling a Multisource (CMDB 360) query, you can create a Multisource (CMDB 360) report that integrates the query results with the platform Reporting feature. You can for example, include such Multisource (CMDB 360) report in the platform CMDB dashboards.

Before you begin

The Multisource (CMDB 360) query for the report must be already saved, have a schedule, and must have already run at least once.

Role required: cmdb_ms_user

About this task

Creating a report that is based on a Multisource (CMDB 360) query, creates a report source which you can then manage using Reporting capabilities.

> **Note:** If you are using the CMDB 360 view in CMDB Workspace to generate the CMDB 360 query and the report, you can skip to step 4 in the procedure below.

Procedure

1. Navigate to **All > Configuration > Multisource Report Builder**.
2. In the Multisource Report Builder list view, select the query that you want to create a schedule for.
3. On the Multisource Report form, click **Create Report**.
4. On the Create a report form, click **Save** or **Run**.

A report shows the results for the most recent query run. If meanwhile the query has changed, then the report shows results that are out of synchronization with the query. When you update the query, ensure to immediately run the updated query so that the report is synchronized with the query.

What to do next

To add a Multisource (CMDB 360) report to the CMDB Correctness Dashboard for example, see Add a report to a dashboard.

Sample CMDB 360/Multisource CMDB queries

Use sample queries to create your own CMDB 360/Multisource CMDB queries.

Discrepancy in multiple attributes between multiple discovery sources

Field	Setting
Name	Discrepancy in multiple attributes between multiple discovery sources (discovery source vs. discovery source)
Description	Find Linux servers with name containing “backup” which have discrepancy in Disk Capacity OR CPU Count OR Serial Number between discovery sources ServiceNow/ServiceWatch/SCCM/Tivoli.
Result type	Multisource data records
Only show difference	Selected
Type of difference	Between discovery sources
Class	Linux Server [cmdb_ci_linux_server]

Field	Setting
Conditions	[Name] [contains] [backup]
Discovery Source	ServiceNow/ServiceWatch/SCCM/ Tivoli
Field to compare	Disk Capacity OR CPU Count OR Serial Number
Limit results to	All

Discrepancy in multiple attributes between CMDB record and discovery sources

Field	Setting
Name	Discrepancy in multiple attributes between multiple discovery sources (CMDB vs. discovery sources)
Description	Find Linux Servers with name containing “backup” which have discrepancy in Disk Capacity AND CPU Count AND Fully Qualified Domain Name for discovery sources ServiceNow/ ServiceWatch/SCCM
Result type	Multisource data records
Only show difference	Selected
Type of difference	Between CMDB record and discovery source
Class	Linux Server [cmdb_ci_linux_server]
Conditions	[Name] [contains] [backup]
Discovery Source	ServiceNow/ServiceWatch/SCCM

Field	Setting
Field to compare	Disk Capacity AND CPU Count AND Fully Qualified Domain Name
Limit results to	All

Servers discovered by ServiceNow but not by Tivoli

Field	Setting
Name	Missing Discovery by Tivoli
Description	Servers discovered by ServiceNow but not by Tivoli
Result type	CI records
Class	Server [cmdb_ci_server]
Discovery Source	[is] [ServiceNow] [is not] [Tivoli]
Limit results to	All

All discovery sources for backup servers

Field	Setting
Name	Discovery Sources Backup Servers
Description	All discovery sources for backup servers
Result type	Data sources
Class	Server [cmdb_ci_server]

Field	Setting
	and class condition: [Host name][starts with][backup]
Limit results to	All

All CMDB 360/Multisource CMDB records where the reported value of Location is different between Altiris and Tivoli discovery sources

Field	Setting
Name	Compare Location-Altiris vs. Tivoli
Description	List all Multisource CMDB records where the reported value of Location is different between Altiris and Tivoli discovery sources
Result type	Multisource data records
Only show difference	Selected
Type of difference	Between discovery sources
Discovery Source	• Altiris • Tivoli
Field to compare	Location
Limit results to	All

All CMDB 360/Multisource CMDB records for Linux Server, where the Location value is different than the reported value by Tivoli

Field	Setting
Name	Linux Server Location - Diff than Tivoli value
Description	All Multisource CMDB records for Linux Server, where the Location value is different than the value reported by Tivoli.
Result type	Multisource data records
Class	Linux Server
Only show difference	Selected
Type of difference	Between CMDB record and discovery source
Discovery Source	[is][Tivoli]
Field to compare	Location
Limit results to	All

Components related to CMDB 360

Several types of components are related to CMDB 360 (included in the com.snc.cmdb plugin), such as tables and properties.

Properties

Open the CMDB 360 Properties page by navigating to **All > Configuration > CMDB 360 Properties**. You can hover over the '?' icon for a property, to show property names.

You must have the cmdb_ms_admin role to modify property values.

Property	Description
glide.identification_engine.multisource_enabled	Enables CMDB 360. • Type: true | false • Default value: false • Location: CMDB 360 Properties page
glide.identification_engine.multisource_cmdb_ci_enabled	Enables capturing CMDB 360 data for CIs from CMDB classes (derived from the cmdb_ci class). • Type: true | false • Default value: true • Location: CMDB 360 Properties page
glide.identification_engine.multisource_non_cmdb_ci_enabled	Enables capturing CMDB 360 data for CIs from non-CMDB classes (not derived from the cmdb_ci class). For example, the Serial Number [cmdb_serial_number] class, or the Software instance [cmdb_software_instance] class. • Type: true | false • Default value: false • Location: CMDB 360 Properties page

Property	Description
glide.identification_engine.multisource.query.batch.limit	Max number of items to show per query results page, in the CMDB 360 Report Builder. Changing the value of this property, might affect performance when running a query. • Type: numeric • Default value: 100 • Location: CMDB 360 Properties page
glide.identification_engine.multisource.query.max.limit	Max number of query results to show when you click **Load All Results** in the CMDB 360 Report Builder. Changing the value of this property, might affect performance when running a query. • Type: numeric • Default value: 10000 • Location: CMDB 360 Properties page
glide.identification_engine.multisource.recompute.max.ci.limit	Max number of CIs that can be included in a CMDB 360 recompute operation. • Type: numeric • Default value: 100000

Property	Description
	• Location: CMDB 360 Properties page
glide.cmdb.logger.source.cmdb_multisource	Enable logging for CMDB 360. CMDB 360 logs are stored in the Log [syslog] table with source set to "cmdb_multisource". • Type: string • Values: info, warn, error, debug, or debugVerbose • Location: Add to System Properties [sys_properties] table.

sn_cmdb_ws.ms.calculate_cmdb_only	Limits the scope of data in the Discovery sources tile in the CMDB 360 dashboard in CMDB Workspace, to CMDB classes only (derived from the cmdb_ci class).
	• Type: true | false
	• Default value: true
	• Values:
	  ◦ True: Include data only from CMDB classes.
	  ◦ False: Include data from CMDB and non-CMDB tables.
	• Location: Add to System Properties [sys_properties]

Property	Description
sn_cmdb_ws.ms.report_class_ci_count_max_threshold	Threshold number of multisource records that when exceeded for a class, CMDB 360 Coverage charts in CMDB Workspace stop populating data for that class (CMDB or non-CMDB tables). When the specified threshold number is reached for a class, the system creates a record in the CMDB 360 Analytics Skipped Class [sn_cmdb_ws_ms_skip_class] table causing that class to be skipped in future job runs. **Note:** Managing class inclusion in the Coverage charts, by using this system property and the CMDB 360 Analytics Skipped Class [sn_cmdb_ws_ms_skip_class] table, can greatly decrease or increase performance of the Coverage chart job. • Type: numeric • Default value: 10 million • Location: Add to System Properties [sys_properties] • Learn more: CMDB 360 experience in CMDB Workspace

Tables

Table	Description
CMDB 360 Data [cmdb_multisource_data]	CMDB 360 data store. Contains the raw data sent by all discovery sources.
CMDB MultiSource Column Metadata [cmdb_multisource_column_metadata]	Mapping of attributes for each class to floatable columns. Used to improves performance of queries that involve high volumes of data.
CMDB Multisource Queries [cmdb_multisource_query]	CMDB 360 query definitions created by the user in CMDB Workspace or in the legacy Multisource Report Builder.
Query Status [cmdb_multisource_query_status]	State of execution, of queries created in CMDB Workspace or in the legacy Multisource Report Builder.
CMDB Multisource Query Results [cmdb_multisource_query_result]	Results for queries created in CMDB Workspace or in the legacy Multisource Report Builder, configured with result type of CI records.
CMDB Multisource Query Result Multisource Records [cmdb_multisource_query_result_ms_record]	Results for queries created in CMDB Workspace or in the legacy Multisource Report Builder, configured with result type of CMDB 360 records.
CMDB Multisource Query Result Discovery Sources	Results for queries created in CMDB Workspace or in the legacy Multisource Report Builder,

Table	Description
[cmdb_multisource_query_result_disco_source]	configured with result type of Discovery source records.
CMDB MultiSource Recompute Task CIs [cmdb_multisource_recomp_task_ci]	All CIs that are involved in a recompute operation.
CMDB Multisource Recompute Tasks [cmdb_multisource_recomp_task]	Recomputation requests and progress status.
CMDB 360 denied classes [cmdb_multisource_deny_class]	Classes that are excluded from CMDB 360 data collection and processing. For any class with an active record, there won't be any data in the CMDB 360 Data [cmdb_multisource_data] table.
CMDB 360 Analytics Skipped Class [sn_cmdb_ws_ms_skip_class]	Internally used to store records of classes for which the threshold number of multisource records, as set in the sn_cmdb_ws.ms.report_class_ci_count_max_threshold property, has exceeded. Classes in this table are excluded from the Coverage charts in CMDB Workspace. You can set the Override class CI records population column for an excluded class to 'true', to include it in Coverage charts future computations.

Table	Description
	Note: Managing class inclusion in the Coverage charts, by using this table and the sn_cmdb_ws.ms.report_class_ci_count_max_threshold system property, can greatly decrease or increase performance of the Coverage chart job.

Roles

Role	Description
cmdb_ms_read	Can access and run a CMDB 360 query but can't create a query. Contains cmdb_read role.
cmdb_ms_editor	Can create and run a query, has full read and write access, but can't do Recompute. Contains cmdb_ms_read role.
cmdb_ms_admin	Can create and run a query, and can modify CMDB 360 properties. Contains cmdb_ms_write role.